

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech CST 3rd Semester Mid-semester Examinations, September 2023

Data Structures CS-2103

Full Marks: 30

Time: 2 hours

Attempt all questions

1. Answer any 4 from the following questions

- Explain why the statement, "The running time of a particular algorithm is at least $O(n^2)$ " is meaningless.
- What do you mean by Abstract Data Type (ADT)?
- How can you create a 2D array of size $M \times N$ using dynamic memory allocation in C programming language?
- What is Memory leak? How can we avoid?
- What are the merits and demerits of a linked list over array?
- What is a Segmentation fault error?

(4x2.5 = 10)

2. Answer any 2 from the following questions

- Let's assume that a set of integers is stored in an array in sorted order. Write a C subroutine `insertAnElement(...)` to insert an element into the array so that the order of elements remain intact.
- Write down the recursive subroutine to find largest among n integers. Then trace the function calls for $n=4$.
- Suppose the running time analysis of a particular algorithm results in $(3n + 8)$ primitive operations. Explain why complexity of the algorithm is $O(n)$?

(2x4 = 8)

3. Answer any 2 from the following questions using C programming language

- Suppose there is a linked list containing integer element. Write a subroutine to remove the last node of the linked list and return its element.
- Define the structure for a stack. Develop the stack subroutines `createStack(...)`, `push(...)` and `pop(...)`.
- A given string consisting of the characters '(' and ')' with other characters. For example, the string, "(b c b (f k))" has parenthesis matched, whereas the string "(b c (b (f k)) a x" does not. Write a subroutine to confirm that the string has matching parenthesis or not using stack (*no need to write stack related functions*).

(2x6 = 12)